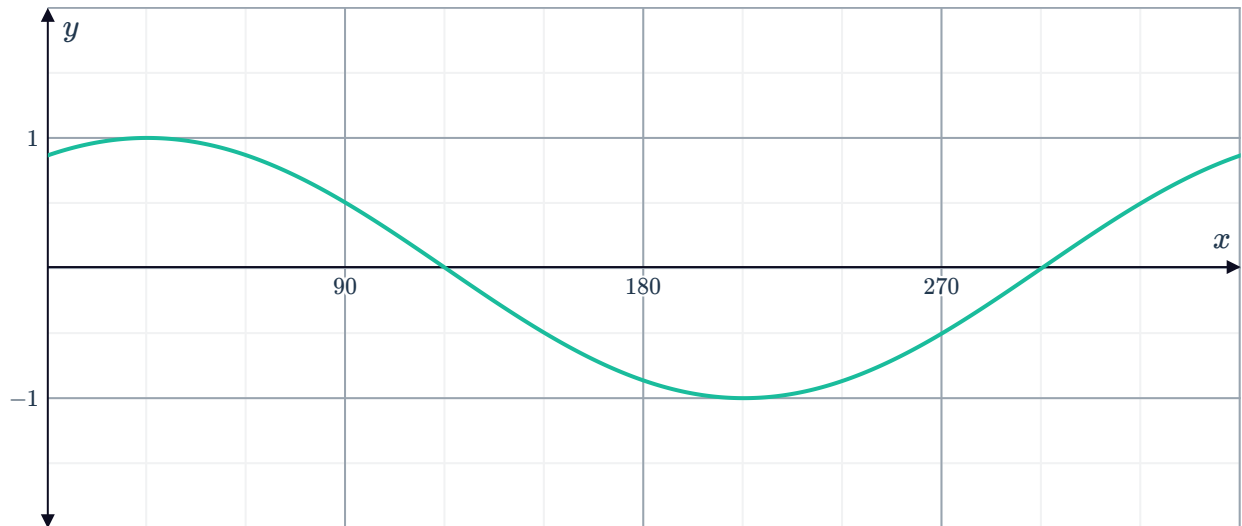


Graphical solutions

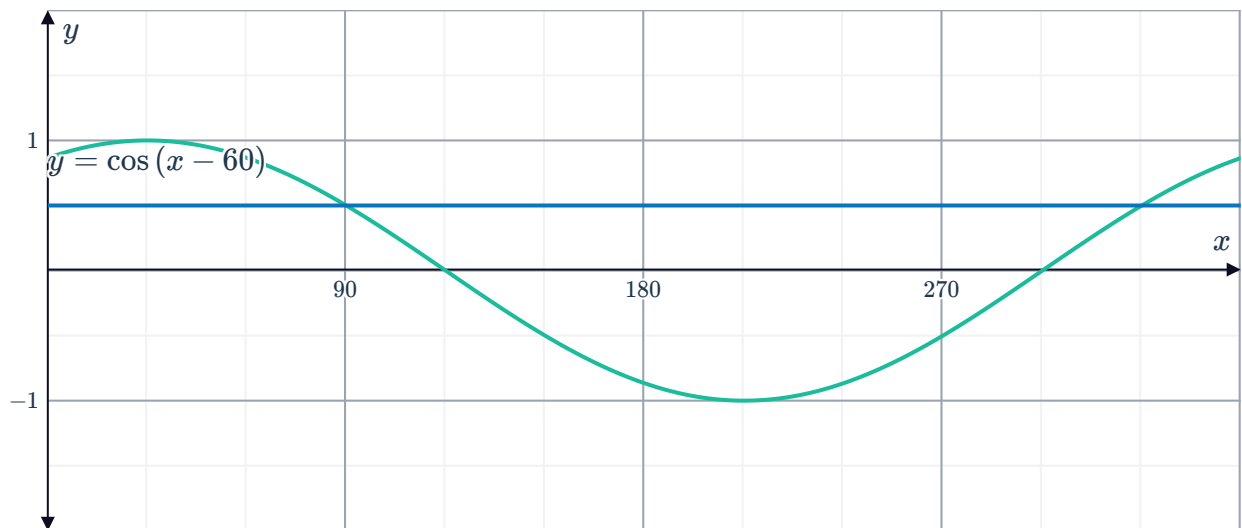


1

a



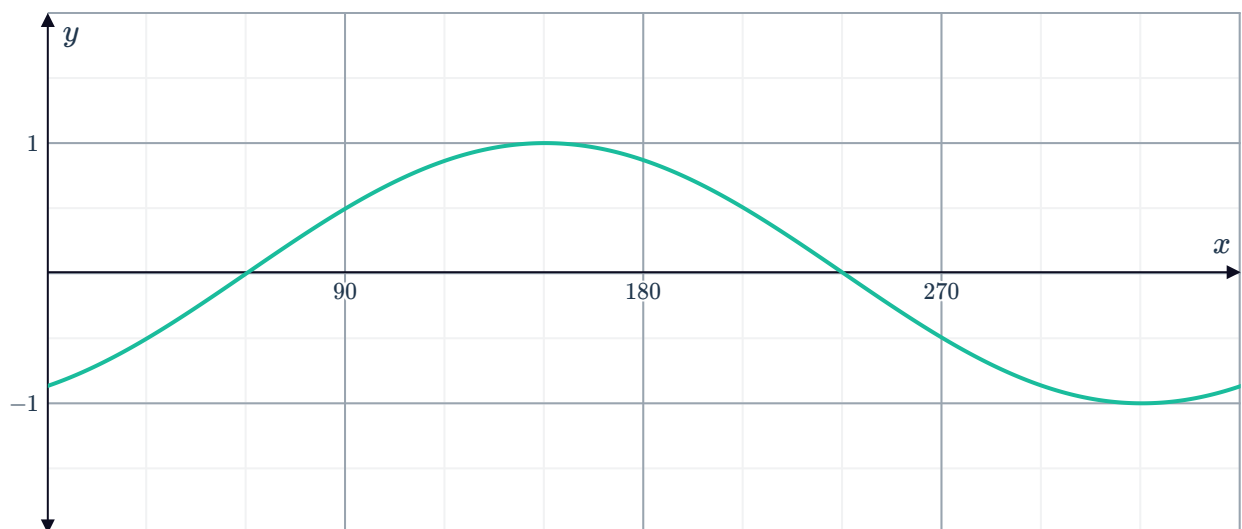
b



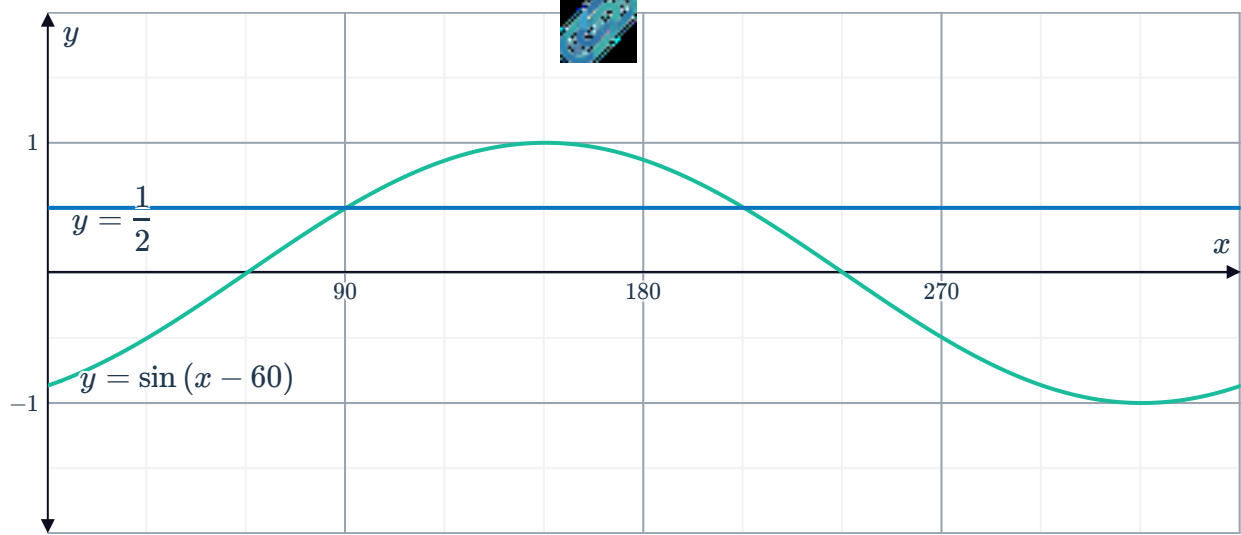
c $x = 90,330$

2

a

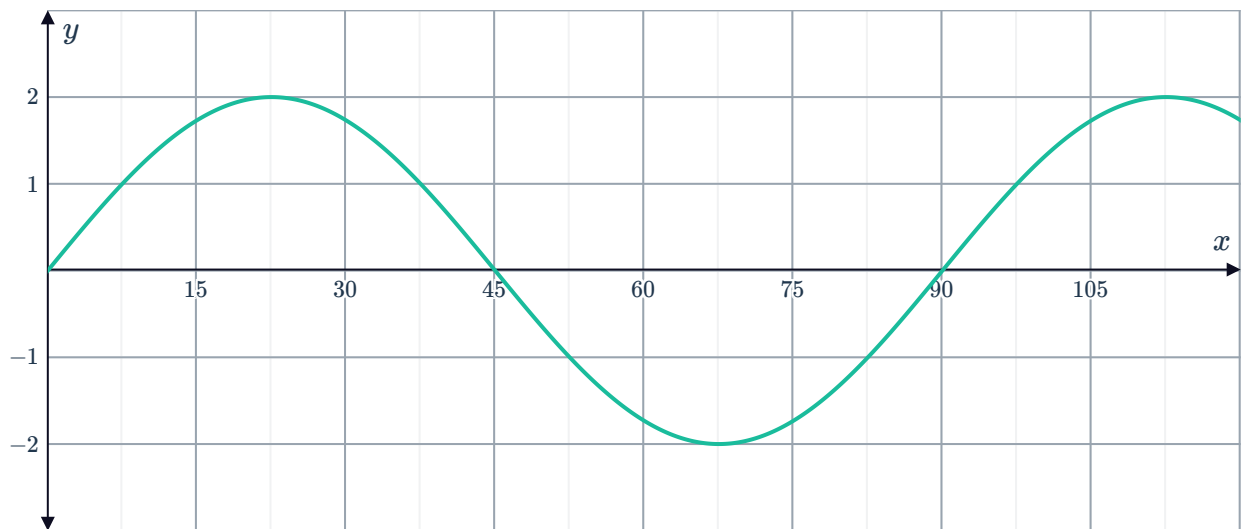


b

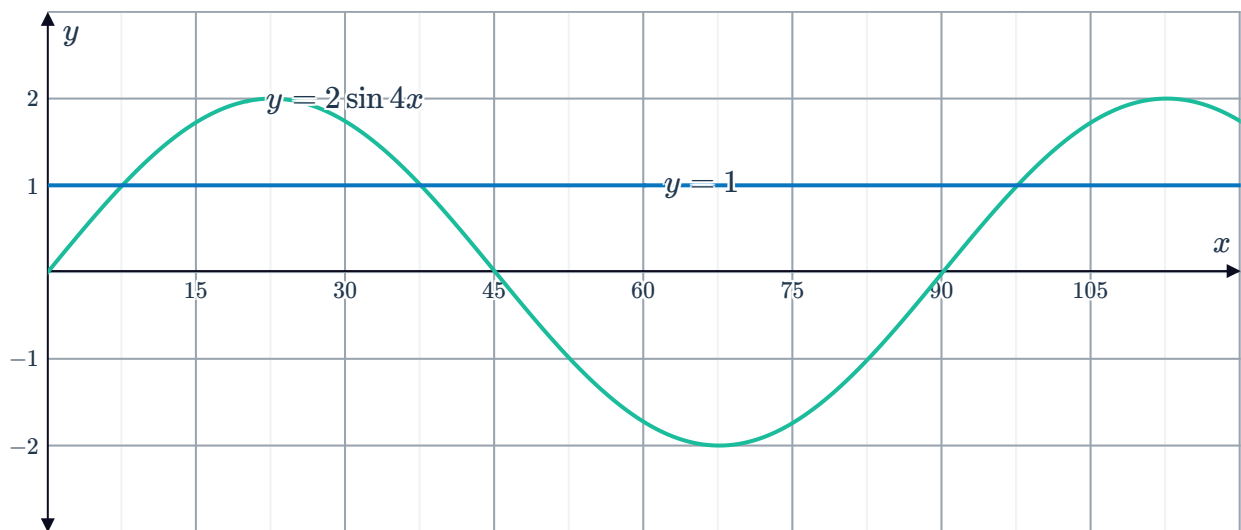
c $x = 90, 210$

3

a



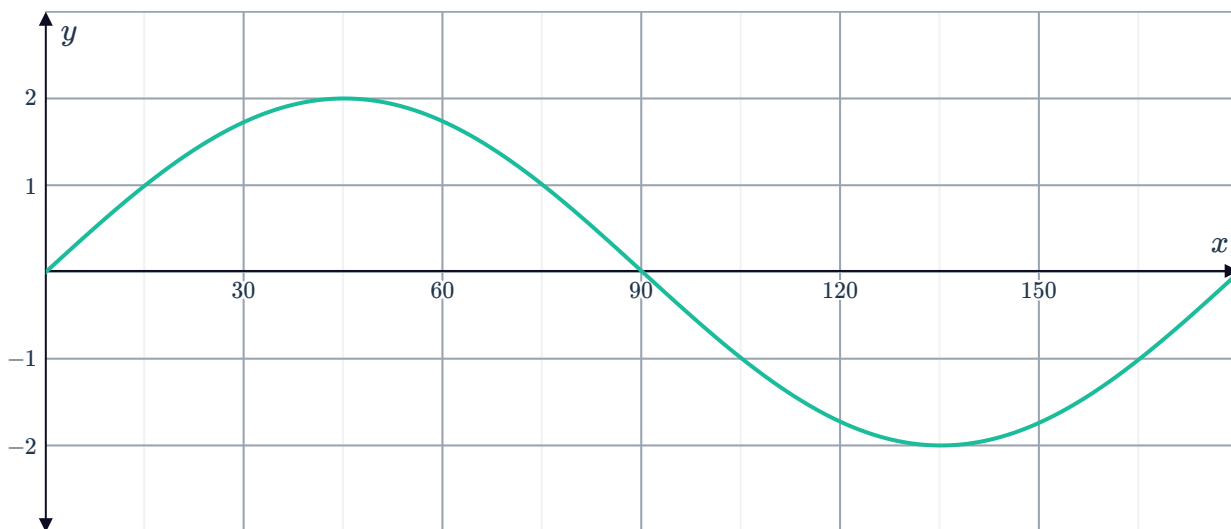
b

c $x = 7.5, 37.5$

4

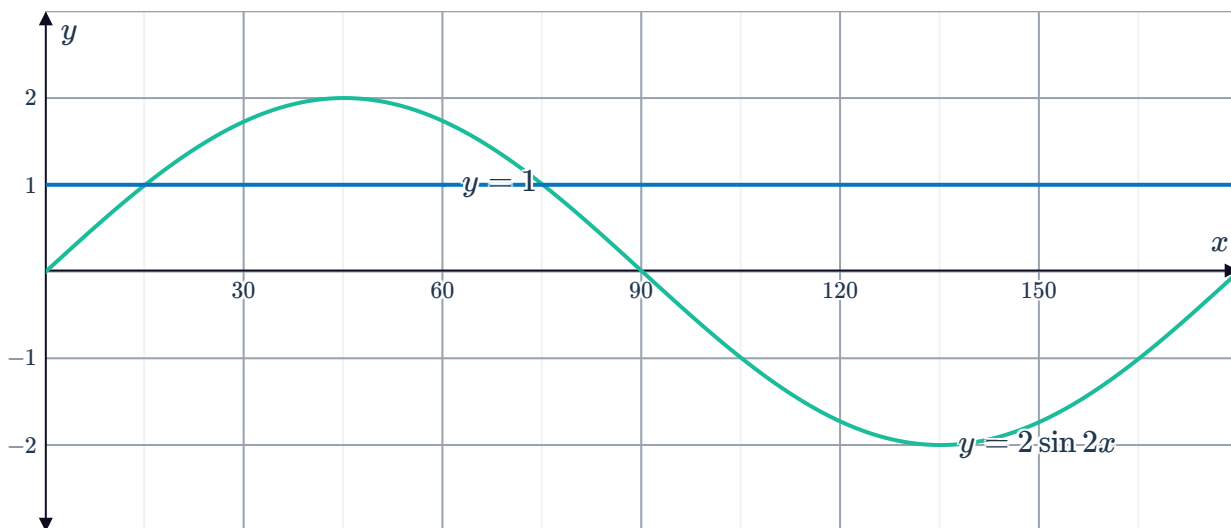


a



b $y = 1$

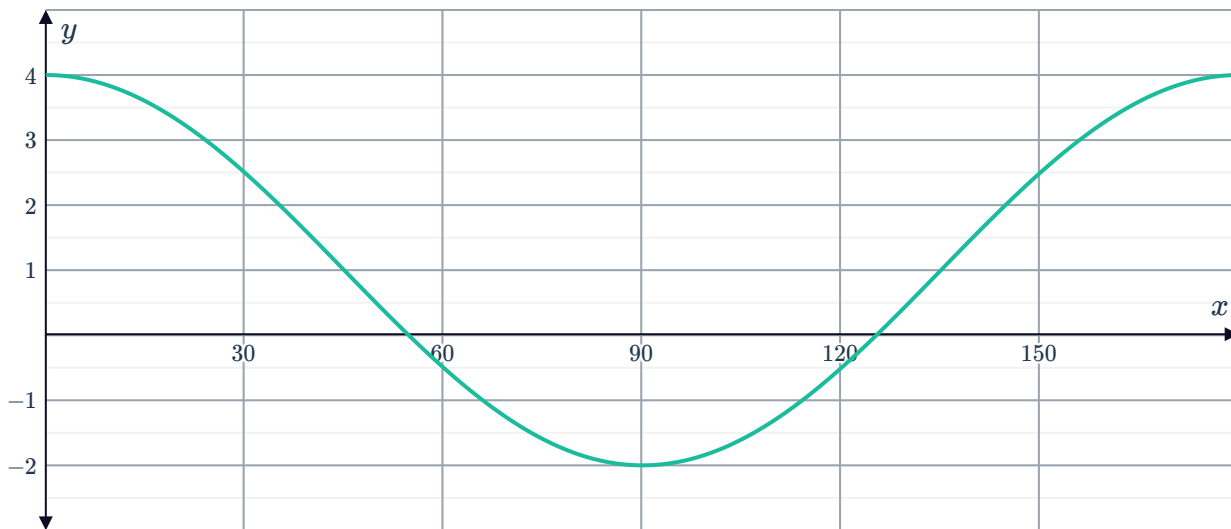
c



d $x = 15, 75$

5

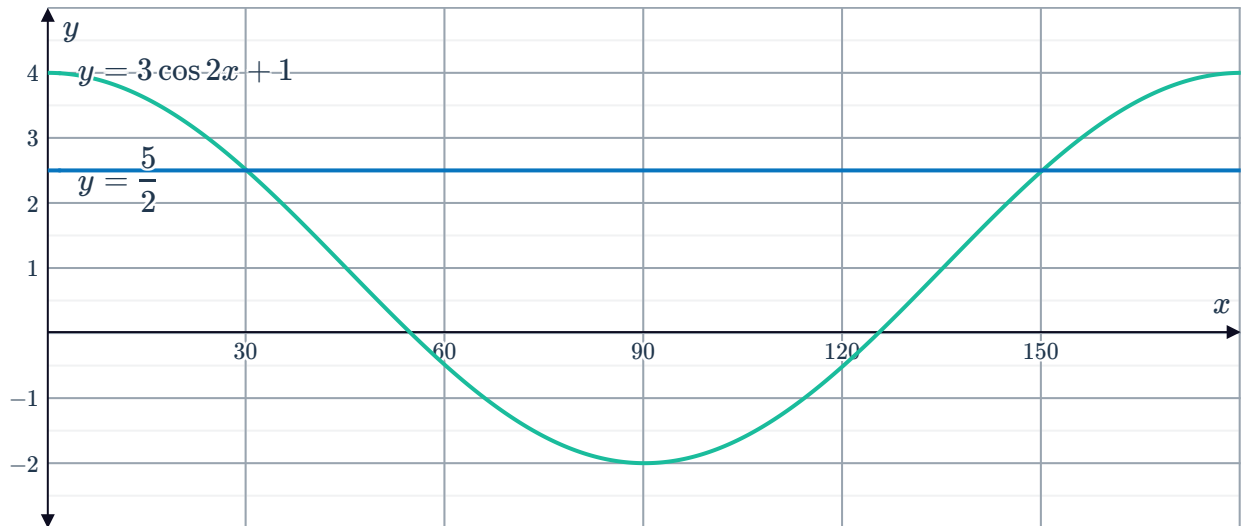
a



b $y = \frac{5}{2}$



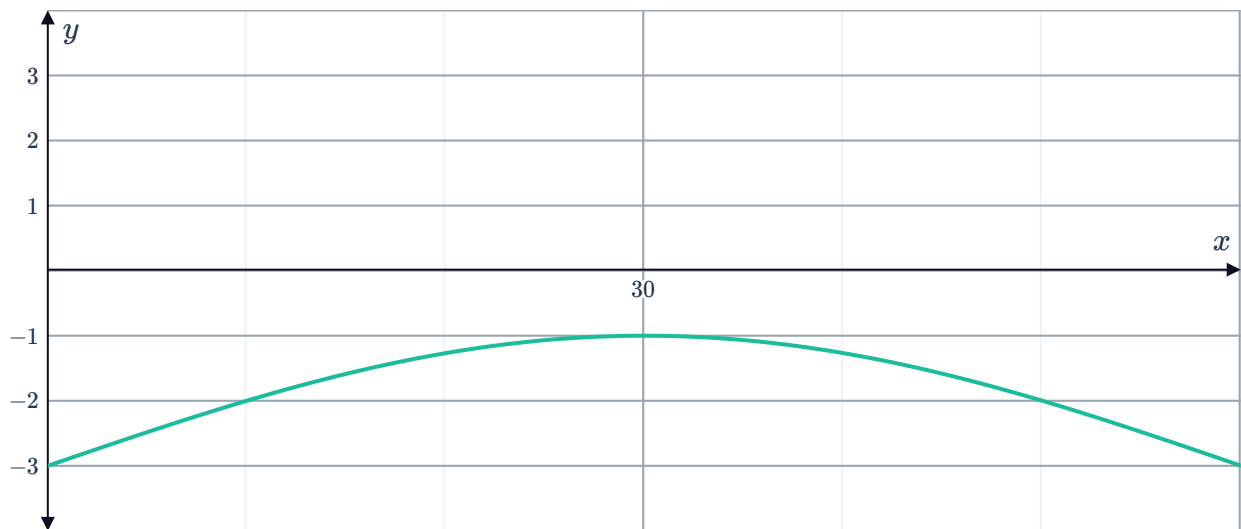
c



d $x = 30, 150$

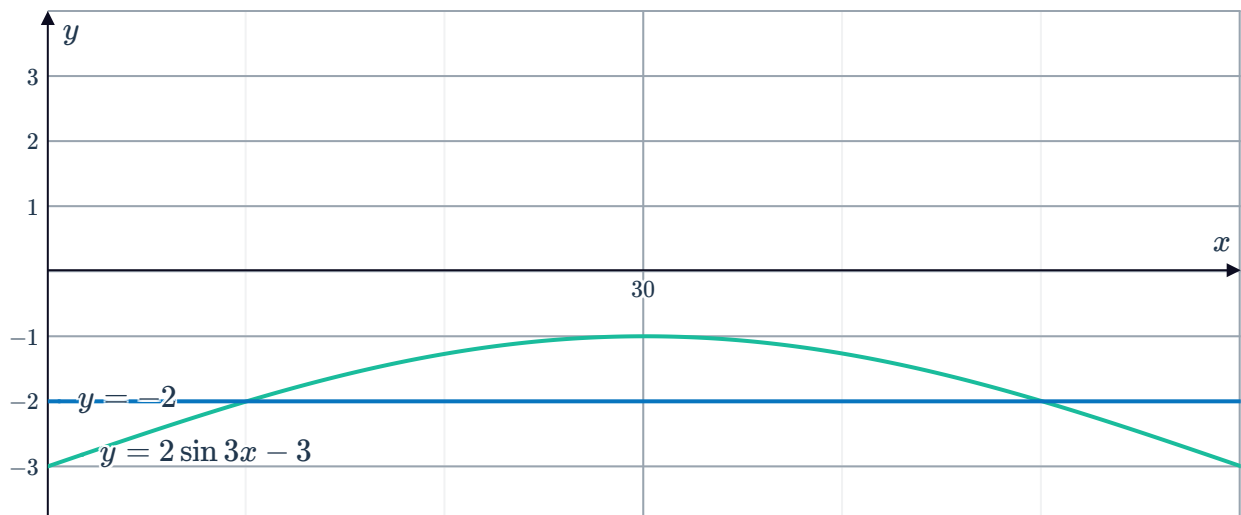
6

a



b $y = -2$

c

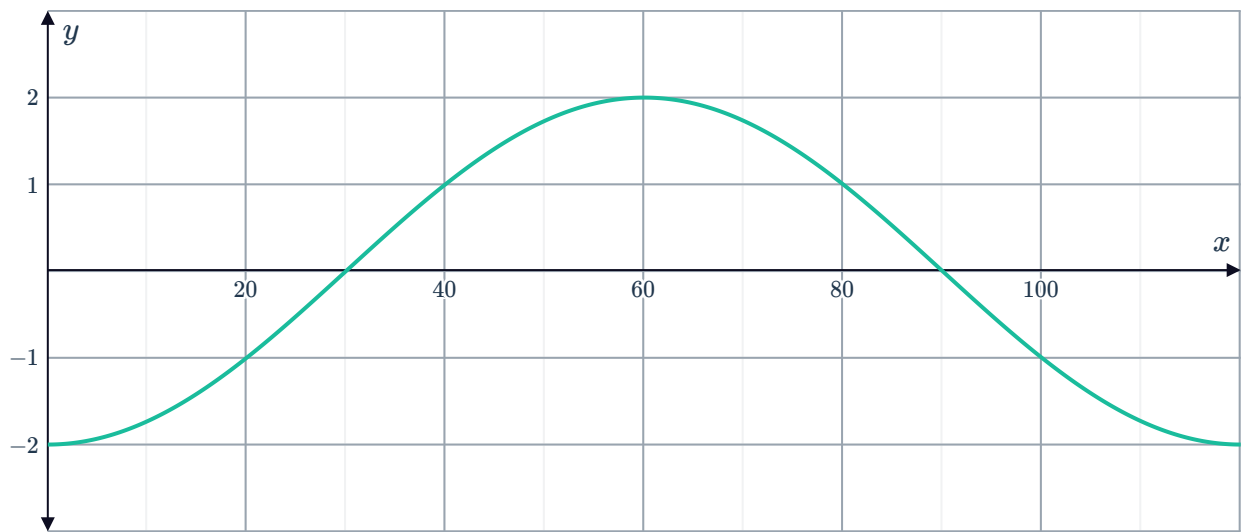


d $x = 10, 50$



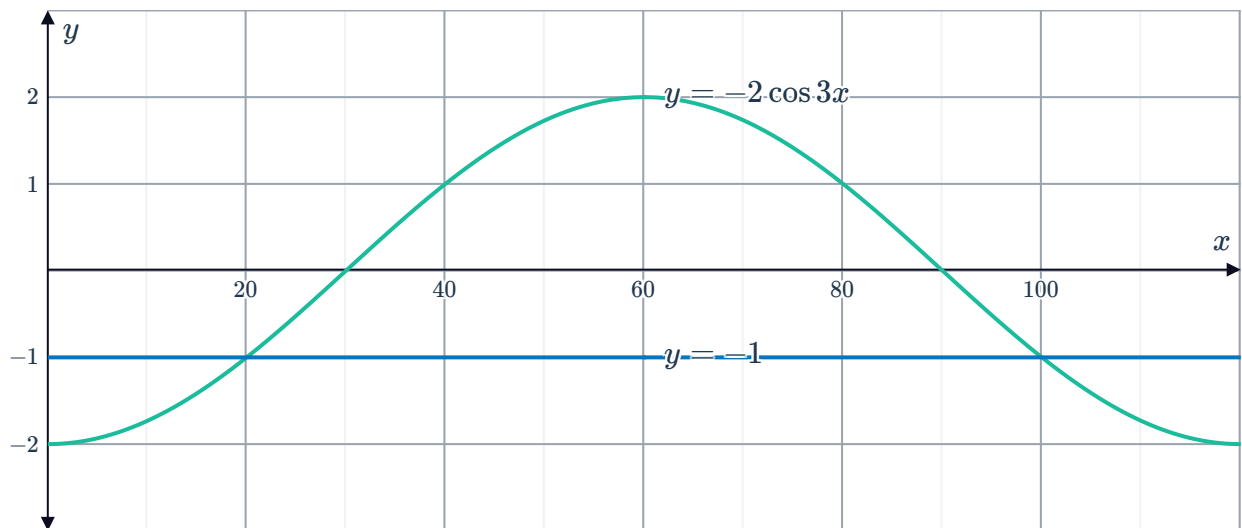
7

a



b $y = -1$

c



d $x = 20, 100$

Exact value equations

8

a No

b Yes

9

a Zero

b Zero

c Zero

10



11

a $\theta = 135^\circ, 225^\circ$

c $\theta = 90^\circ, 270^\circ$

e $\theta = 180^\circ$

g $\theta = 135^\circ, 225^\circ$

i $\theta = 90^\circ$

k $\theta = 0^\circ, 180^\circ, 360^\circ$

m $\theta = 135^\circ, 315^\circ$

o $\theta = 120^\circ, 240^\circ$

b $\theta = 60^\circ, 300^\circ$

d $\theta = 30^\circ, 150^\circ$

f $\theta = 225^\circ, 315^\circ$

h $\theta = 240^\circ, 300^\circ$

j $\theta = 60^\circ, 240^\circ$

l $\theta = 150^\circ, 330^\circ$

n $\theta = 60^\circ, 300^\circ$

p $\theta = 45^\circ, 135^\circ$

Non-exact value equations

12

a $\theta = 37^\circ$

b $\theta = 40^\circ$

c $\theta = 36^\circ$

d $\theta = 21^\circ$

e $\theta = 66^\circ$

13

a $\theta = 25^\circ, 335^\circ$

b $\theta = 143^\circ, 217^\circ$

c $\theta = 220^\circ, 320^\circ$

d $\theta = 69^\circ, 111^\circ$

e $\theta = 35^\circ, 215^\circ$

f $\theta = 144^\circ, 324^\circ$